

An Historical Sketch of The Conceptions of Memory among the Ancients.

**by
William A. Burnham,**

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Author: William Henry Burnham

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HISTORICAL SKETCH OF THE CONCEPTIONS OF MEMORY
AMONG THE ANCIENTS ***

**An
Historical Sketch
of
The Conceptions of Memory among the
Ancients.**

Submitted as a Thesis
by
William H. Burnham,
Candidate for the degree of Doctor of Philosophy
at
Johns Hopkins University,
Baltimore.
1888.

Table of Contents.

I.

Conceptions of Memory before Aristotle, pp. [1](#)–11

II.

Aristotle's Conceptions of Memory, pp. [12](#)–40

III.

Conceptions of Memory among the Stoics and Epicureans, and in Cicero and Quintilian, pp. [41](#)–46

IV.

Conceptions of Plotinus and St. Augustine, pp. [47](#)–70

V.

Diseases of Memory mentioned by ancient writers, pp. [71](#)–73

VI.

Ancient Systems of Mnemonics,

pp. [74](#)–
76

Memory.

I.

Mnemosyne, Hesiod tells us, was the mother of the Muses. Without speculating as some have done about the reasons for this myth it is interesting as showing an appreciation of the fundamental nature of memory and some sort of crude introspective psychology dating back possibly to pre-historic times.

Before the art of writing was in common use men had to depend more largely than to-day upon their memories for preserving and transmitting their knowledge. It is not surprising, therefore, that the ancients put a high estimate upon memory before they began to theorize about its nature. There are, of course, allusions to memory in Homer and in the Hebrew Scriptures. ^[1] And occasionally one of the early Greek philosophers tries to explain some phenomenon of memory. But we find no scientific study of the subject before Aristotle.

The psychology of the Ionian school of philosophers, as far as they can be said to have had any at all, was sensationalism. Their views of memory must be conjectured from the fundamental principles of their philosophy.

The doctrine of transmigration as held in the Pythagoreans is an anticipation of Plato's doctrine of reminiscence, but there is little psychology in it. Theophrastus tells us that Diogenes of Appollonia was puzzled by the phenomenon of forgetting things. ^[2] But he explained it in accordance with the principles of his philosophy by supposing that the cause of forgetting was an arrest of the equal distribution of air throughout the body. A corroboration of this explanation he found in the easier breathing that follows the recalling of what was forgotten.

Among the Eleatics, Parmenides is reported to have held that not only thought, but recollecting and forgetting depended upon the way the light or heat and the dark or cold are mixed in the body. If we may trust

Theophrastus,^[3] it may be assumed that, according to Parmenides, every presentation corresponded to a definite mixture or relation of these qualities, and with the destruction of that relation the presentation disappeared, that is, was forgotten.

Heraclitus, one might suppose, would study memory carefully, but in the fragments of his philosophy that have come down to us nothing is said upon the subject.

In Plato we find a more modern psychology. According to him the thinking power of the mind, the understanding, is above the mere power of sense perceptions. It is this power which compares and considers, notes similarities and contrasts, unity and plurality, and forms ideas of relation between Being and Non-Being as well as relations of number and proportions. Among the elements of this power, recollection (ἀνάμνησις) is of prime importance. This rests upon the association by similarity and simultaneity.^[4]

Plato distinguishes the passive retention (μνήμη) of perceptions from active memory (ἀνάμνησις),^[5] and suggests as a definition of memory, “the power which the soul has of recovering, when by itself, some feeling which it experienced when in company with the body.” He attempts no explanation of memory ~~of memory~~; but in the Theaetetus puts the following words into the mouth of Socrates:

“I would have you imagine, then, that there exists in the mind of man a block of wax, which is of different sizes in different men; harder, moister, and having more or less of purity in one than in another, and in some of an intermediate quality.... Let us say that this tablet is a gift of Memory, the mother of the Muses, and that when we wish to remember anything which we have seen or heard, or thought in our own minds, we hold the wax to the perceptions and thoughts, and in that receive the impressions from them as from the seal of a ring; and that we remember and know what is imprinted as long as the image lasts; but when the image is effaced, or cannot be taken, then we forget and do not know.”

Plato carries out the same figure to explain different degrees of memory. “When the wax in one’s soul is deep, abundant, smooth, and of the right quality, the impressions are lasting. Such minds can easily retain and are not liable to confusion. But, on the other hand, when the wax is very soft, one learns easily and forgets as easily; if the wax is hard, the reverse is true;

again, if the wax is hard or impure, the impressions are indistinct; and still more indistinct are they when jostled together in a little soul.”^[6]

This illustration must not be taken too seriously; for later on in the same dialogue Socrates calls it a “waxen figment” and substitutes for it the figure of the aviary of all kinds of birds—“some flocking together apart from the rest, others in small groups, others solitary, flying anywhere and everywhere.”

The receptacle is empty when we are young. The birds are kinds of knowledge. Learning is the process of capturing the birds and of detaining them in this enclosure. In acts of memory we re-catch them and take them out of the aviary.

Plato’s views upon memory have a special interest on account of their connection with his metaphysical doctrines. Perception and recollection are the occasion of the minds turning away from the world of sense to the inner world of innate and universal ideas. These ideas we could never get from sense-perception. That gives us only the immediate and the individual. The ideas are of the essential and the universal. We could not conceive them if we did not already know them. Hence the power to know the universal in the individual proves a previous existence in which we had the intuitions of universal truths; and, accordingly, learning is but recollection.^[7] The metaphysical aspects of memory, however, let us avoid as much as possible. They would soon lead far from a psychological study. But this doctrine of recollection lies at the heart of the Platonic philosophy, and it is necessary to note carefully the distinction between this and ordinary memory. The latter, as defined by Plato in the passage quoted above is the memory or recollection of what has been learned through the body, that is, through sense-perception, belongs to the world of appearances, and is liable to many errors. The former, on the other hand, is not concerned with things of sense. It is recollection of that higher world where we had an antenatal vision of intelligible realities. Its highest manifestation is the insight of the philosopher who sees the divine goodness, truth, and beauty.^[8]

II.

The difference in philosophic method between Plato and Aristotle is well illustrated by their treatment of memory. What Plato says of memory is incidental to the discussion of such profound matters as the nature of the soul and the theory of knowledge. Memory, according to him, is one of the higher faculties and partakes of the eternal nature of the soul. Aristotle makes a special study of memory in a less transcendental way. With him it is no longer a function of the eternal Nous; but it has its seat in the passive Reason, is dependent upon a physiological process, and perishes with the body.

Aristotle seems to have been the first of ancient philosophers to write a systematic treatise on psychology. But, rather curiously, in this work on psychology there is no special treatment of memory. A special tract, however, was devoted to the subject.^[9] This, so far as we know, was the first scientific study of memory; and for this reason, as well as for its intrinsic merits, the tract deserves special attention. But before passing to his doctrine of memory, it is well to notice briefly his theory of sense-perceptions. On occasion of appropriate stimuli movements take place in the sense-organs. These movements, however, are not sense perception. In perception the mind must compare and distinguish disparate sensations; it must unite the sensations presented simultaneously by our double sense organs as of sight and hearing, and it must be conscious of sensation. This work of comparison, of psychic synthesis, and of self-conscious perception is performed by a central sense. The physical basis of this sense is the heart. Through it the mind performs the act of sense perception. The functions now attributed to nervous substance are referred by Aristotle to the pneuma connected with the blood. This is the medium by which the movements arising in the sense organs are transmitted to the heart, and in this pneuma

the movements persist long after the external stimuli have ceased to act. Incidentally, it is interesting to note, that according to Aristotle's psychology, the brain has very little to do with mental activity. To borrow a phrase from Wallace, it serves simply as "a cooling apparatus to counteract the excessive warmth of the heart." When the movement occasioned in the sense organ by an external stimulus is propagated to the heart,^[10] it becomes a perception of the soul. Sense perception, then, is an act of the soul by means of a physiological process. In the words of Aristotle it is "a movement of the soul through the body."^[11] Now this movement sometimes continues after the stimulus, which was the occasion of it, has ceased to act. The extreme case is the well known phenomenon of a visual after-image. The images of the imagination are such after-sensations. Imagination is weak sensation, or in the words of Hobbes, "decaying sense". So too, dreaming is the result of a movement in our bodily organs, caused either from without or from within. Again, these persisting movements are the elements of memory.

At first, one wonders how Aristotle will distinguish those movements which constitute memory from those which are the basis of imagination. He is not entirely satisfactory on this point; but he makes the following distinction. The picture of the imagination, or the corresponding movement does not refer to an external object, and is not located in the past. The memory picture, on the other hand, does refer to an object and carries with it the consciousness of a time in the past, when the perception remembered took place.^[12] Memory, then, involves time, and both this and the sense of time are dependent upon the central sense. Memory, as we have seen, is dependent upon the residua of sensations. The subjective side of a sensation is an image. Thus memory belongs to the same part of the soul as the imagination, and the proper objects of memory are images (φανταστά). The image is a condition (πρόθος) of the central sense. Memory per se is of the original image or perception, and only in an accidental manner does it relate to matters of thought.^[13]

In his special tract on memory, Aristotle in part repeats Plato's views, in part discusses the obvious facts of memory, which, having been continually repeated since his time, have become mere platitudes, and in part he tries to explain the phenomena of memory in accordance with his general system of psychology. The essay, however, is of special interest, because in it Aristotle sets forth very clearly the famous doctrine of association of ideas. Some of

the other points in the treatise may be briefly mentioned and special consideration given to the portion relating to recollection and association. ¶ First Aristotle takes considerable space to show what would seem to be apparent enough to everybody, that memory is of the past, as perception is of the present, and hope and opinion of the future.

The central sense or sensorium must be in a condition suitable to receive and retain impressions. If the sensorium is too hard, no impression is made. If it is greatly agitated, the new movement is ineffectual: on somewhat the same principle, one may suppose, as one say in modern psychology, that a weak stimulus is washed out by a strong one. Hence the very young and the very old have poor memories; for the former are in the movement of growth, the latter in that of decay. Again, the question arises: How is it that in recollection we recognize the memory image as a picture of the absent object? A scholastic answer is given. “An animal painted in a picture, he says, is both an animal and a copy, and while being thus one and the same, it is nevertheless two things at once. The animal and the copy are not identical, and we may think of the picture either as animal or as a representation. This is also true of the image within us; and the idea which the mind contemplates is something, although it is also the image of something else.”^[14]

The second chapter of the treatise on memory is devoted chiefly to recollection and the association of ideas. Aristotle distinguishes carefully ~~between~~ the mere persistence and reproduction of a presentation (μνήμη) from voluntary recollection (ἀνάμνησις). The latter is indirect reproduction. It is possible only by the association of ideas. The former is an attribute of animals, while the latter is peculiar to man. Recollection occurs according to the sequence of ideas.^[15] What and how necessary the sequence shall be depends upon our past experience. “If the sequence be necessary”, Aristotle continues, “then, when this movement occurs, that one will follow. If it is not necessary, but a matter of habit, the latter movement will generally follow.” Sir W^m. Hamilton understands the word translated movement (κίνησις) to mean merely change in quality. The word, then, he thinks may be fairly translated into modern nomenclature by his famous term modification. One hesitates to criticise such a profound scholar and such a diligent student of Aristotle as Sir W^m. Hamilton; but in the light of what has been said it seems much simpler and more in accordance with the psychology of Aristotle to understand his doctrine of recollection as

follows: The physiological movements originally connected with a series of perceptions must occur again in the same order when we recall a true memory-picture.^[16] Man is so constituted that when one movement and the mental image connected with it occur, another movement with its appropriate mental image is likely to follow. When we would recall anything, therefore, we must call up idea after idea until we arrive at one upon which the one we are in search of has often been sequent in our experience. Or in terms of physiology, movement after movement must recur until we arrive at a movement upon which the movement corresponding to the idea desired has often been sequent.

This sequence or association of ideas is subject to certain laws. The remarkable passage in which Aristotle states these laws is translated by Sir W^m. Hamilton as follows:

“When, therefore, we accomplish an act of Reminiscence we pass through a certain series of precursive movements until we arrive at a movement, on which the one we are in quest of is habitually consequent. Hence, too, it is that we hunt through the mental train, excogitating [what we seek] from [its concomitant in] the present, or some other time and from its Similar or Contrary or Coadjacent. Through this process Reminiscence is effected. For the movements, [which and by which we recollect] are, in these cases, ~~and~~ sometimes the same, sometimes at the same time, sometimes parts of the same whole: so that [having obtained from one or the other of these a commencement], the subsequent movement is already more than half accomplished.”^[17]

Wallace quotes the same passage in the introduction to his Psychology of Aristotle,^[18] and gives the following somewhat different and probably more accurate translation:—

“When engaged in recollection we seek to excite some of our previous movements until we come to that which the movement or impression of which we are in search was wont to follow. And hence we seek to reach this preceding impression by starting in our thought from an object present to us or something else whether it be similar, contrary, or contiguous to that of which we are in search; recollection taking place in this manner because the movements are in one case identical, in another case coincident and in the last case partly overlap.”^[19]

Whichever translation we adopt it seems plain enough that Aristotle maintained that voluntary recollection depended upon the laws of association by similarity, contrast or contrariety, and contiguity. Very likely he meant to include simultaneity and sequence; but any proof of this should rest upon the general import of the passage rather than upon any doubtful emendation like Hamilton's.^[20]

A more important question is whether Aristotle meant to limit the application of these laws to voluntary recollection (ἀνάμνησις), or whether he intended to include spontaneous reproduction (μνήμη) as well.

The opinion commonly held by students of Aristotle, from Themistius down, has been that he applied the law of association only to voluntary recollection. Hamilton, however, argues forcibly that Aristotle taught the universality of the law of association. It seems natural enough to suppose that one who saw so clearly that in the voluntary train of thought the sequence conforms to the law of association, would have seen that the same laws apply to the spontaneous activity of the mind. But while Aristotle states the law of association clearly for the former, he at most merely alludes to the latter, and obscurely enough at that. Later in the same treatise Aristotle gives an illustration that may serve to elucidate the principles of association that have just been stated. In recollection there are certain movements which serve as standpoints or clues. Milk suggests whiteness, whiteness the clear atmosphere, the atmosphere moisture, this the rainy season. So too, Themistius in commenting upon the passage quoted above, uses an illustration somewhat similar. "I see a painted lyre, and moved by this, as the prior and leading image, I have the reminiscence of a real lyre; this suggests the musician, and the musician, the song I heard him play."^[21] Again, Aristotle uses an illustration somewhat as follows: Let A. B. C. D. E. F. G. H. represent a series of ideas, one of which we will to recall. From D E, as a starting point we may be moved forward by E or backward by D, by the association of ideas. If, then, on the suggestion of D E, we do not find what we would recall, we may find it by running over the series E ... H; if not, we shall at any rate find the desired idea by running over the series backward from D to A. Not much stress, however, should be put upon this last illustration; for the text is so obscure that many different interpretations have been given by commentators. Perhaps Aristotle meant to illustrate something more profound than the mere linkings of presentations in a series, and the process of recollecting the mental train. But the illustration

of such a simple matter as this was not unimportant in the first scientific study of memory.

The place of memory in the Aristotelian psychology in relation to the lower psychic activities is plain from what has been said. The relation of memory, as voluntary recollection, to the higher activity of the Nous is indicated by Aristotle when he says that recollection is a syllogistic process. Thus it is that, while many animals have the lower kind of memory, man alone has the higher form. "The reason is", says Aristotle, "that Reminiscence is, as it were, a kind of syllogism, or mental discourse. For he who is reminiscent, that he has formerly seen or heard, or otherwise perceived, anything virtually performs an act of syllogism."^[22] With Aristotle the higher functions of the soul are based upon the lower. "Without nutrition, there is no sense; without sense there is no phantasy; without phantasy there is no cogitation or intelligence."^[23] The place of memory among the soul's functions is, with the phantasy or imagination, mediate between sensation and intelligence.

In connection with Aristotle's doctrine of recollection, one passage in his Psychology is interesting, although its importance has, perhaps, been exaggerated. "Recollection," he says, "starts from the soul and terminates in the movements or impressions which are stored up in the organs of sense."^[24] Siebeck interprets this passage as meaning that the soul has the power by means of the heart to effect a sort of efferent movement towards the sense organs and thus to arouse anew the persisting residua of former motions. Recollection, then, with Aristotle as in modern psychology, is an excitation of the sense organs, reproduced in a less degree; and the same organs are excited and the same movements repeated as in the original sensation.^[25] This passage is certainly a remarkable anticipation of Bain's famous doctrine that a reproduced impression "occupies the very same parts and in the same manner" as the original impression.^[26]

In the foregoing sketch of Aristotle's view of memory the attempt has been made to give only what can fairly be found in Aristotle's text. Much of his tract upon memory is obscure. Commentators have held very conflicting opinions in regard to the importance of what he wrote upon association and recollection. Sir W^m. Hamilton calls him "the founder and finisher of the theory of Association," looks upon the commentators as marvellously stupid in their interpretations, and deems it a proof of Aristotle's genius that

it took the world 2000 years to become intelligent enough to understand him. Indeed, in reading Hamilton's erudite discussion one may be led almost to believing that Aristotle was the first Scottish philosopher. But while Hamilton's Scottish apperception probably found too much in Aristotle's treatise, and while, on the other hand, Lewes may be right in saying that "here as in so many other cases modern knowledge supplies the telescope with its lenses", nevertheless Aristotle's doctrine of association was a valuable contribution to science. And it is manifestly unfair to charge him with ignorance of its importance, because he did not spin out as many volumes upon the subject as the English associationists have done.^[27]

III.

The Stoics took Plato's figure of the wax almost literally! They held that the mind is originally a tabula rasa. Sensations are the first writing upon this tablet. The object of sensation makes an impression upon the perceiving subject, as the seal impresses the wax. Memory depends upon this impression. This was the view of Zeno. Chrysippus found difficulties in such a crude materialistic theory. How could the mind receive and retain at the same time a number of different and partly incompatible impressions? Accordingly he replaced this view by the theory that the sense impression consists in a qualitative change (ἀλλοίωσις) of the passively receiving organ, the soul.^[28] The presentation (φαντασία) is a state of the soul. The relation of memory to the general theory of knowledge with the Stoics was briefly as follows:—The lowest act of the soul is mere perception (αἴσθησις); the next is presentation (φαντασία), which adds conscious observation, its function being to make a first test of the truth of the material furnished by sense. If perception has offered a true picture of the external object, this presenting activity of the mind becomes so intensive that the understanding is brought into action. The understanding or judgment approves or disapproves the presentations. If it approves, there arises the empirical fact, which bears upon it the mark of truth. These facts memory stores up. By combination of the separate facts empirical concepts are formed which make up the treasure of memory or experience.^[29]

The psychology of Epicurus and the other atomists was a simple kind of mechanical sensationalism. Eidola or images from external objects enter the soul through the sense organs. The mind stores up a great multitude of these eidola. Whenever we call up a picture of memory or of the imagination, we turn the attention to one of these images. Thus the mind sees in the same

way that the eye does, with this difference, that it perceives much thinner eidola.^[30]

Cicero and Quintilian both dwell upon the importance of memory; and both seem to adopt the common theory of the time, that impressions are stamped on the mind as the signets are marked on wax. They are especially concerned, however, with principles relating to the exercise of memory; and they give instructions for mnemonic aids in oratory. Cicero lays special stress upon order as an aid to memory; and as sight is the most acute of the senses, those things are best remembered which are visualized by the imagination. In accordance with the ancient mnemonic systems he would have these imagined forms localized. The advice of Quintilian in respect to memory is especially sensible. According to him, nothing can take the place of exercise and labor. Next in importance is the division and arrangement of one's subject. He notices also the importance of good health; and says that for slow minds an interval of rest is a good thing, though he seems to be uncertain whether the advantage is due to the rest, or whether it gives reminiscence time to mature.^[31]

IV.

The Neo-Platonic psychology of memory is represented by Plotinus.^[32] He discusses the subject at considerable length, and presents a somewhat original doctrine. Memory does not belong to God nor to the divine immutable intelligence in man, which knows by direct intellectual perception. It is a function of the soul and first appears when the world soul is individualized in bodies. Memory, however, has no basis in the physical organism, nor does the soul impress the sensations upon the body. The effects of sensations are not like impressions made by a seal, nor are they reactions (ἀντερείσεις), or configurations (τυπώσεις), but in sense-perception as in thought the soul is active. In memory, too, the soul is active, not passive. The influence of the body proves nothing against this. The changeable nature of the body may cause us to forget, but it cannot condition positive recollection. The body is the river of Lethe, but memory belongs to the soul. The part of the soul to which memory belongs is the image-forming faculty. This holds sense impressions as well as thought. Two souls, the higher and the lower, are concerned in memory. When the soul leaves the body, the recollections of the lower soul are soon forgotten in proportion as the higher soul rises toward the intelligible world.^[33]

St. Augustine developed the views of the Neo-Platonists in regard to memory. With him memory is a faculty of animals, men, and angels. God, whose immutable essence is above the sphere of movement and change, does not remember. Everything is seen by him in one indivisible and unchangeable present.

Augustine does not agree with Aristotle that some animals are devoid of memory. He attributes memory even to fishes, and relates in confirmation of this opinion an incident that he had observed. There was a large fountain filled with fishes. People came daily to see them and often fed them. The

fishes remembered what they received; and as soon as any one came to the fountain they crowded together expecting their accustomed food. But Augustine does not suppose that animals have that higher memory which is purely intellectual, although he probably failed to see how purely mechanical and involuntary their so-called acts of memory are. Memory with St. Augustine, as in the psychology of Plotinus, is a function of the soul, not of the body. But with Aristotle he refers it to the central sense.^[34]

What is memory? It is thinking of what one knows. All the various modifications of the soul cannot all be present to us at once. There is a difference between knowing a thing and thinking of it. The musician, says Augustine, knows music, but he does not think of it when he is talking about Geometry.^[35] The ideas relating to music are in the mind in a latent state. Augustine anticipates Leibnitz in discussing the unconscious modifications of our ~~own~~ ideas; but he speaks especially of their gradual decay, while Leibnitz considers the unconscious growth of them. “Many numbers”,^[36] Augustine says, “are gradually effaced from memory; for they remain not an instant unaltered. Indeed what is not found in memory after a year is somewhat diminished even after one day. But this diminution is imperceptible; yet it is not wrongly inferred; for it does not suddenly all vanish the day before the year is up. Hence we may conclude that from the moment it was engraved in memory it began to slip away.”^[37] This doctrine of unconscious mental changes and unconscious mental states is one of the most remarkable features of Augustine’s psychology. With irresistible logic he demonstrates the existence of such states in the following passage from another place:—

“But what when the memory itself loses anything, as falls out when we forget and seek that we may recollect? Where, in the end, do we search, but in the memory itself? And there, if one thing be, perchance, offered instead of another, we reject it, until what we seek meets us; and when it doth we say ‘This is it’; which we should not unless we recognized it, nor recognize it, unless we remembered it——For we do not believe it as something new, but upon recollection, allow what was named to be right. But were it utterly blotted out of the mind, we should not remember it, even when reminded. For we have not as yet utterly forgotten that which we remember ourselves to have forgotten. What, then, we have utterly forgotten, though lost, we cannot even seek after.”^[38] It would not be difficult to find passages in

modern psychologies that read almost like translations of this chapter of Augustine's Confessions.

Two kinds of memory—sense memory and intellectual memory are distinguished in the Augustinian psychology. The former preserves and reproduces not only the images of visible objects, but also the impressions of sounds, odors and other objects which strike our senses.^[39] The images are not like the eidola of Democritus, but are ideal, formed by the mind from its own essence. Intellectual memory contains our knowledge of the sciences, of literature, and dialectic, and of the questions relating to these subjects.^[40] This memory, unlike the memory of sense, contains not the images of things, but the things themselves. These ideas which the intellectual memory stores up are in a sense innate. They never came to us through the senses. They could never have been taught to us, unless we had already had them in our memories. “When I learned them I gave not credit to another man's mind, but recognized them in mine.” Thus the memory contains the idea of truth and of God.

Augustine points out, too, what has been repeated by Locke and others until it has become a platitude, that we do not remember objects themselves, but the ideas which we have gained from them. And with his usual subtlety he shows that much of what is ordinarily attributed to perception is really the work of memory.

“We see what importance St. Augustine attaches to memory. It is in his view the faculty which preserves the ideas relating not only to the body, but to the soul, not only to eternal truths, but to the Eternal Being himself——. This memory, which is peculiar to man, and which animals do not possess—this memory, which in a mysterious manner contains in it intelligible realities, is, according to the Bishop of Hippo, one of the three great faculties of man, and the origin of the other two. It is from it that intelligence arises, and the will proceeds from the one to the other and unites them. Thus, if it is allowed to compare things human with things divine, we have in us an image of the august Trinity. Memory, in which is the matter of knowledge, and which is as the place of intelligible things, offers some resemblance to the Father; the intellect, which is derived ~~from~~ and formed from it, is not without analogy to the Son; and love or will, which unites the intelligible (or memory) to the intellect has a certain resemblance to the Holy Spirit.”^[41] The phenomena of memory were also

important with Augustine as weapons against materialism. By memory the soul knows objects of sense when it no longer perceives them, and, moreover, combines the heterogeneous in ways inexplicable by means of a physical substance. And again the soul can form abstract conceptions of space and mathematical truths.

The well-known conditions of a good memory, such as acuteness of sensation, order, and repetition, Augustine notices only incidentally. More attention is given to the relation of the will to memory and to the association of ideas.

Whether we remember or not depends upon the will. By an act of will we avert the memory from sense-perceptions, as, for example, when we hear a speaker and do not notice what he says, or read a page and do not know what we have read, or walk with our attention upon something else. In all these cases we perceive, but do not remember our perceptions. So, too, recollection depends upon the will: “As the will applies the sense to the body (or external object), so it applies the memory to the sense, and the eye of the mind of the thinker to the memory.”^[42]

This power of the will over memory is, however, limited by the association of ideas. In order to recall anything by a voluntary effort we must remember the general notion of the thing or some associated idea. “For example, if I wish to remember what I supped on yesterday, either I have already remembered that I did sup, or if not yet this, at least, I have remembered something about that time itself, if nothing else; at all events, I have remembered yesterday and that part of yesterday in which people usually sup, and what supping is.”^[43] In another place he says that, of a series of ideas the lost part is recovered “by the part whereof we had hold.”

Many since Augustine have marvelled at the miracle of memory. None have expressed their admiration more eloquently. “Great is this force of memory”, he exclaims, “excessive great, O my God; a large and boundless chamber; who ever sounded the bottom thereof? Yet is this a power of mine, and belongs unto my nature; nor do I myself comprehend all that I am. Therefore is the mind too strait to contain itself. And where should that be which it containeth, not of itself? Is it without and not within? How then doth it not comprehend itself. A wonderful admiration surprises me, amazement seizes me upon this. And men go abroad to admire the heights of mountains, the mighty billows of the sea, the broad tides of rivers, the

compass of the ocean, and the circuits of the stars, and pass themselves by; nor wonder that when I spake of all these things, I did not see them with my eyes, yet could not have spoken of them, unless I then actually saw the mountains, billows, rivers, stars, which I had seen, and that ocean which I believe to be inwardly in my memory, and that, with the vast spaces between, as if I saw them abroad. Yet did not I by seeing draw them into myself, when with mine eyes I beheld them; nor are they themselves with me, but their images only. And I know by what sense of the body, each was impressed upon me.”

It is an interesting fact that Augustine noticed the possibility of illusions of memory. Certain rare phenomena—the so-called recollections of Pythagoras and others who were said to have remembered objects perceived in a former state of existence—he explains in a very modern fashion, except that he attributes these beliefs to the agency of evil spirits. “For we must not”, he says, “acquiesce in their story who assert that the Samian Pythagoras recollected some things of this kind, which he had experienced when he was previously here in another body; and others tell yet of others, that they experienced something of the same sort in their minds. But it may be conjectured that these were untrue recollections, such as we commonly experience in sleep, when we fancy we remember, as though we had done or seen it, when we never did or saw at all; and that the minds of these persons, even though awake, were affected in this way at the suggestion of malicious and deceitful spirits, whose care it is to confirm, or to sow some false belief concerning the changes of souls, in order to deceive men.”^[44] If they truly remembered such things, he argues, such phenomena would not be as rare; but many persons would experience the same.

Perhaps the most serious criticism of Augustine’s psychology of memory is that he entirely neglects the physiological side of the subject. He does not even notice the relation of memory to states of health or disease, and of youth or age. In one place, however, he states that memory has its seat in one of the three ventricles of the brain, which is situated between that which is the seat of sensation and that which presides over ~~of~~ locomotion, so that our movements may be coordinated.^[45]

The criticism has also been made that Augustine seems to waver in his conception of memory, that he sometimes represents it as the source of all our intellectual activity comparing it among the other faculties to the Father

in the Trinity; that again he seems to limit this faculty to the work of preserving knowledge acquired empirically. Certainly in some passages he seems to make memory contain a kind of innate ideas that may be drawn forth by suggestion.^[46]

But if Augustine is unsatisfactory in this, it must be remembered that he is not writing a psychology and that he was, as Ferraz calls him, a philosopher of transition. “He combats Plato’s doctrine of reminiscence, and prepares the way for the innate ideas of Descartes, without positively enough rejecting the former, and without clearly enough admitting the latter.”^[47]

V.

The pathological side of memory seems to have been little studied by the ancients. Augustine referred to the possibility of illusions of memory in the way already mentioned. Seneca tells of a certain Sabinus who had so bad a memory that he forgot the name of Ulysses, and again of Achilles, and sometimes of Priam, though he knew them as well as we remember our schoolmasters.^[48] Some remarkable cases of amnesia were reported to the Elder Pliny. “Nothing whatever in man,” he says, “is of so frail a nature as the memory; for it is affected by disease, by injuries, and even by fright; being sometimes partially lost and at other times entirely so. A man who received a blow from a stone forgot the names of the letters (of the alphabet) only; while, on the other hand, another person, who fell from a very high roof could not so much as recollect his mother or his relations and neighbors. Another person in consequence of some disease forgot his own servants even; and Messala Corvinus, the orator, lost all recollection of his own name.”^[49] While these cases are good illustrations of certain diseases of memory, they are not reported with sufficient accuracy and detail to render them of much scientific value. Ancient thinkers appear not to have seen the importance of studying the pathological conditions of memory.

VI.

No historical sketch of memory among the ancient Greeks and Romans is complete without some mention of their mnemonic systems. The art of mnemonics seems to have been much in vogue among them. There are frequent allusions to this art in the works of Aristotle, Plato, and other classic writers. Aristotle is reported by some to have written a work upon mnemonics. Every scholar of the classics is familiar with the story that ascribes the invention of the art to Simonides.

The main principles of the ancient mnemonic systems according to Cicero and Quintilian were as follows. The thing to be remembered was localized by the imagination in some definite place—say in a room of a real or imaginary house; and, if necessary, a concrete symbol as vivid as possible was associated with it. This method was used by the Romans as an aid in oratory; and it has been said that the phrases, “in the first place,” “in the second place,” and the like, originated in this ancient practice.

The ancient systems of mnemonics are inferior to the best modern systems, that, since the days of Pick^[50] have been based upon sound psychological principles. But the ancient systems were probably very helpful to eye-minded people. The men with remarkable memories, mentioned by Cicero and other ancient writers very likely owed much to mnemonic aids. It is of special psychological interest to consider the ancient mnemonic devices in the light of such studies as those of Galton upon mental imagery, number forms, and the like.^[51] The high estimate that many of the ancients placed upon the mnemonic art, may, perhaps, fairly be taken as evidence that what Galton calls the faculty of visualisation was developed among them. Especially some of the Roman orators seem to have possessed this faculty in a high degree.

Biographical Note of the Author.

I was born at Dunbarton, N.H. on the 3d of Dec. 1855, and am the youngest son of Samuel and Hannah Dane Burnham. I graduated at the High School at Manchester, N.H. in 1875. The next three years I spent in teaching and in study. In 1878 I entered Harvard College, and graduated in the Class of '82. The following year was spent in teaching in the Preparatory Department of Wittenberg College. The next two years were spent at the State Normal School at Potsdam, N.Y., where my work was the teaching of Latin and Rhetoric. In 1885 I entered the Johns Hopkins University. In this university I have held the positions of Fellow in Philosophy and Fellow by Courtesy. My work has been chiefly under the direction of Prof. G. Stanley Hall and Dr. Richard T. Ely.

William Henry Burnham.

FOOTNOTES:

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1. For reference see Carus: *Geschichte der Psychologie*, pp. 150 & 169.
 2. Theophrastus, 45.
 3. Theophrastus, 4. Cf. Siebeck: *Geschichte der Psychologie*. Erste Abtheilung, p. 150.
 4. Phaedrus, 73.
 5. Philebus, 34.
 6. Jowett's Translation.
 7. For references see Zeller's *Plato and the Older Academy*, pp. 126 and 407. Cf. also Siebeck: *Geschichte der Psychologie*.
 8. For the many passages in which the words *μνήμη*, *μνημονεύω*, *μνημονικός*, *μῆμων*, *μνήμων* occur in Plato, ~~conf~~ see *Ast. Lexicon Platonicum*, II. pp. 356–357. For *ἀναμνήσκω* and *ἀνάμνησις*

cf. the same, Vol. I. pp. 151–152.

[9.](#) De Memoria et Reminiscentia. For a list of commentaries, see Hamilton's edition of Reid's Works, p. 891.

[10.](#) Touch and Taste according to A. reside in the heart. Sight, Sound, and Smell in the brain, but they are indirectly connected with the heart.

[11.](#) De Somno. 1. 454.

ἡ δὲ λεγομένη αἴσθησις ὡς ἐνέργεια κινήσις τις διὰ τοῦ σώματος τῆς ψυχῆς ἐστίν.

[12.](#) See Wallace; Aristotle's Psychology, Introduction, pp. 93 & 94.

[13.](#) In other words, abstract ideas and the like are reproducible only so far as they imply images.

[14.](#) Quoted from G. H. Lewes's work on Aristotle, p. 257.

[15.](#) Συμβαίνουνσι δ' αἱ ἀναμνήσεις, ἐπειδὴ πέφυκεν ἡ κίνησις ἥδε γενέσθαι μετὰ τήνδε.

This passage is obscure, but it is generally understood to refer to the sequence of motions or the corresponding ideas, and this interpretation agrees with the context. See Hamilton's Edition of Reid, pp. 892–893, and Themistius' paraphrase of De Memoria, quoted by Hamilton, p. 893–894; also Siebeck, Geschichte der Psychologie, Zweite Abtheilung, p. 77; Grote's Aristotle, p. 215; Grant's Aristotle, p. 170; Wallace's Aristotle's Psychology, Introduction, p. 95.

[16.](#) Cf. Siebeck, Geschichte der Psychologie, Zweite Abtheilung, p. 77.

[17.](#) The Greek is as follows:

διὸ καὶ τὸ ἐφεξῆς θηρεύομεν νοήσαντες ἀπὸ τοῦ νῦν ἢ ἄλλου τινὸς καὶ ἀφ' ὁμοίου, ἢ ἐναντίου, ἢ τοῦ σύνεγγυς διὰ τοῦτο γίνεται ἡ ἀνάμνησις. αἱ γὰρ κινήσεις τούτων τῶν μὲν αἱ αὐταί, τῶν δ' ἅμα, τῶν δὲ μέρος ἔχουσιν ὥστε τὸ λοιπὸν μικρὸν ὃ ἐκινήθη μετ' ἐκεῖνο.

[18.](#) p. 95.

[19.](#) See also Grote, Grant, Siebeck, and Zeller: see opp. cit.

[20.](#) Hamilton's emendation is as follows:

After ἢ ἄλλου τινὸς in the passage cited he would supply χρόνου or καιροῦ.

[21.](#) Quoted by Hamilton in his edition of Reid's Works, p. 901.

[22.](#) Hamilton's translation. Edition of Reid's Works, Vol. II. p. 909.

[23.](#) Grote, op. cit. p.

[24.](#) Wallace: Aristotle's Psychology, p. 41.

[25.](#) Siebeck: op. cit., pp. 78–79.

[26.](#) Bain: Senses and Intellect, p. 338. 3d American ed.

[27.](#) For passages where the words *μνήμη*, *ἀνάμνησις*, etc., occur in Aristotle see the Index in the Berlin edition of Aristotle's works, Vol. V. In addition to the works cited see also Waddington-Kastus; *De la psychologie d'Aristote*, Chap. XIII.

[28.](#) For reference see Siebeck's *Geschichte der Psychologie*, p. 209. See also Ueberweg, *History of Philosophy*, Vol. I. p. 193.

[29.](#) Cf. Stein; *Die Erkenntnisstheorie der Stoa*; Zeller; *Stoics, Epicureans, and Sceptics*; and Ueberweg; *Hist. of Philos.* loc. cit.

[30.](#) Lucretius, IV. 75 seqq.

[31.](#) Cicero, *De Oratore*, II. 86 seq.

Quintilian, *Instit. orat.* XI. 2 seq.

[32.](#) Cf. Enn. IV. L. III. C. XXV. XXX and L. VI.

[33.](#) Cf. Siebeck: *Geschichte der Psychologie*, II. p. 314 seq.

[34.](#) The central sense or sensorium, however, according to Augustine is located in the brain, not in the heart as in Aristotle's psychology.

[35.](#) *De Trin.* L. XIV. C. VII. See also Ferraz,—*Psych. de St. Augustin*, 2nd ed.

[36.](#) Augustine does not mean to limit what follows to mathematical truths, but according to his psychology the same would be true of anything that we are liable to forget.

[37.](#) *De Musica*, L. VI. C. IV.

[38.](#) *Conf.* L. X. C. XIX. Pusey's translation.

[39.](#) *Conf.* L. X. C. VIII.

[40.](#) *Conf.* L. X. C. IX. sqq.

[41.](#) Ferraz.—*Psych. de St. Augustin*, p. 178. Cf. also *De Trin.* L. XV. C. XXI. XXII. XXIII. and L. XI. C. VII. and VIII.

[42.](#) *De Trin.* L. XI. C. VIII. Pusey's translation.

[43.](#) *De Trin.* L. XI. C. VII. Pusey's translation.

[44.](#) *De Trin.*, L. XII. 15.

[45.](#) *De Gen. ad Litt.*, L. VII. C. XVIII.

[46.](#) *Conf.*, L. X. C. X. and XI.

[47.](#) Ferraz, *op. cit.*, p. 192.

[48.](#) *Epistolae*, 27.

[49.](#) Nat. Hist., L. VII. C. 24.

[50.](#) Cf. his work “On Memory and the Rational Means of Improving it.” London, 1862.

[51.](#) Galton, Enquiry into Human Faculty, p. 83 seq.



TRANSCRIBER'S NOTES

1. P. [68](#), added a footnote anchor.
2. Silently corrected typographical errors and variations in spelling.
3. Retained anachronistic, non-standard, and uncertain spellings as printed.
4. Footnotes have been re-indexed using numbers and collected together at the end of the last chapter.

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